Exercise sheet: Week 1

1. a) Show that for the Normal distribution, N(μ, σ^2), the sample mean is an unbiased and consistent estimator for μ.

   b) How far can you generalise this result to other distributions whose parameters include their mean μ?

2. A sample of size n is taken from the Beta$(\theta, 1)$ distribution, with p.d.f. $f(x;\theta) = \theta x^{\theta-1}$, $0 < x < 1$. Find the method of moments estimator for θ.

3. Instead of matching the sample mean and variance to model values, an alternative version of the method of moments matches the sample mean and second moment, setting $\sum x_i/n = \mathbf{E}[X]$ and $\sum x_i^2/n = \mathbf{E}[X^2]$. Show that for the Normal distribution N(μ, σ^2) this leads to the variance estimator $\tilde{\sigma}^2 = \sum(X_i - \bar{X})^2/n$.

4. Consider a random sample, $X_1, X_2, ..., X_n \sim$ N$(0, \theta)$. Find a formula for confidence intervals using the pivot $Z^\star = \sum X_i/\sqrt{n\theta}$, and compare with the CI based on the pivot $Z = (\sum X_i)^2/n\theta$.

5. Consider a random sample, $X_1, X_2, ..., X_n \sim$ U$[0, \theta]$. Find a pivot based on $X_{(n)} = \max\{X_i\}_{i=1}^n$, and hence a 95% CI for θ, using equal tails.

   Apply your result to data from a sample of size $n = 5$, for which $x_{(5)} = 1.3$.

   [Hint: for the uniform distribution on (0,1), the jth order statistic $X_{(j)} \sim$ Beta$(j, n + 1 - j)$.]

6. Find a formula for CIs for θ based on a random sample from the Poi(θ) distribution, using the Normal approximation to this distribution.